

Lemma.

Let  $K/F$  be a field extension, and let  $\alpha \in K$  be algebraic over  $F$ . Then, for any  $\sigma \in \text{Aut}(K/F)$ ,  $\sigma\alpha$  is a root of the minimal polynomial for  $\alpha$  over  $F$ .

Proof. Let  $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in F[x]$  be the minimal polynomial for  $\alpha$ . Then,  $f(\alpha) = 0$ , so  $\sigma(f(\alpha)) = 0$ . (The field homomorphism preserves zero).

Now,  $0 = \sigma(f(\alpha)) = \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_1\alpha + a_0)$

$$\begin{aligned} \sigma \text{ fixes } F \quad \hookrightarrow &= (\sigma\alpha)^n + \sigma(a_{n-1}) \cdot (\sigma\alpha)^{n-1} + \dots + \sigma(a_1) \cdot (\sigma\alpha) + \sigma(a_0) \\ &= (\sigma\alpha)^n + a_{n-1}(\sigma\alpha)^{n-1} + \dots + a_1(\sigma\alpha) + a_0 \\ &= f(\sigma\alpha). \end{aligned}$$

Prop. Let  $H \leq \text{Aut}(K)$  be a subgroup of the group of all automorphisms of  $K$ .

Then, the collection  $F$  of elements of  $K$  fixed by all the elements of  $H$  is a subfield.

Proof. First, given  $\sigma \in \text{Aut}(K)$ , the set  $F_\sigma = \{a \in K \mid \sigma(a) = a\}$  is subfield, because:

Let  $a, b \in F_\sigma$  be given. Then,  $\sigma(a-b) = \sigma(a) - \sigma(b) = a-b$ , and if  $b \neq 0$ , then  $\sigma(a \cdot b^{-1}) = \sigma(a) \cdot \sigma(b)^{-1} = a \cdot b^{-1}$ , so the claim is proved.

And now, since the arbitrary intersection of fields is field, so proved. ~~QED~~  
Such a subfield is called the fixed field of  $S \subseteq \text{Aut}(K)$ .

Prop. Let  $K$  be a field. Then,

If  $F_1 \leq F_2 \leq K$ , then  $\text{Aut}(K/F_2) \leq \text{Aut}(K/F_1)$ .

If  $H_1 \leq H_2 \leq \text{Aut}(K)$ , then  $F_{H_2} \leq F_{H_1}$ .

In other say, The inclusion relation of automorphisms and fixed field is reverse.

Lemma.

Def. A finite extension  $K/F$  is called a Galois Extension if:

$$|\text{Aut}(K/F)| = [K:F].$$

In such a case, denote  $\text{Gal}(K/F) = \text{Aut}(K/F)$ .

Def. A character  $\chi$  of a group  $G$  with values in a field  $L$  is a group homomorphism

$$\chi: G \rightarrow L^\times$$

The characters  $\chi_1, \dots, \chi_n$  of  $G$  are said to be linearly independent over  $L$  if:

$$\alpha_1 \chi_1 + \dots + \alpha_n \chi_n \equiv 0 \text{ for some } \alpha_i \in L, \text{ then } \alpha_1 = \dots = \alpha_n = 0.$$

[Thm. [Dedekind-] Independence of Characters Theorem]

If  $\chi_1, \dots, \chi_n: G \rightarrow L^\times$  are distinct characters of  $G$  with values in  $L$ , then they are linearly independent over  $L$ .

Proof. Suppose that the characters are linearly dependent.

That is, there exists non-trivial representation: for some  $\alpha_1, \dots, \alpha_n \in L$ ,

$$\alpha_1 \chi_1 + \dots + \alpha_n \chi_n \equiv 0 \text{ where } \alpha_1, \dots, \alpha_n \text{ are not all zero.}$$

So, the set  $\{m \in \mathbb{N} \mid \text{for some } b_1, \dots, b_m \in L \setminus \{0\}, \text{ and } \sigma \in S_n, b_1 \chi_{\sigma(1)} + \dots + b_m \chi_{\sigma(m)} \equiv 0\}$  is non-empty, thus it has a minimal element by the well-ordering principle.

$m \in \mathbb{N}$  with  $b_1 \chi_1 + \dots + b_m \chi_m \equiv 0$ , the indices are renumbered,  $b_i \in L \setminus \{0\}$ .

Since the characters are distinct,  $\chi_1(g_0) \neq \chi_m(g_0)$ , for some  $g_0 \in G$ .

Given  $g \in G$ , observe that

$$0 = b_1 \chi_1(g_0 g) + \dots + b_m \chi_m(g_0 g) = b_1 \chi_1(g_0) \chi_1(g) + \dots + b_m \chi_m(g_0) \chi_m(g)$$

Meanwhile,

$$0 = \chi_m(g_0) \cdot 0 = \chi_m(g_0) \cdot (b_1 \chi_1(g) + \dots + b_m \chi_m(g)) = b_1 \chi_m(g_0) \chi_1(g) + \dots + b_m \chi_m(g_0) \chi_m(g),$$

Subtracting gives  $[\chi_m(g_0) - \chi_1(g_0)] \cdot b_1 \chi_1(g) + \dots + [\chi_m(g_0) - \chi_{m-1}(g_0)] \cdot b_{m-1} \chi_{m-1}(g) = 0$ , because the last term is terminated. But, since  $[\chi_m(g_0) - \chi_1(g_0)] \neq 0$ , it is another representation, but that is contradiction to the minimality of  $m$ .

[Corollary. Distinct Field Automorphisms are linearly independent.]

[Thm.

Let  $G = \{\sigma_1 (= \text{id}), \sigma_2, \dots, \sigma_n\}$  be a subgroup of  $\text{Aut}(K)$ , and let  $F = K^G$ .

Then,  $[K:F] = |G| = n$ .

Proof. To show the equality, we will show that neither  $|G| > [K:F]$  nor  $|G| < [K:F]$ . Suppose that  $n > [K:F]$  and let  $w_1, \dots, w_m$  be a basis for  $K$  over  $F$ . (So,  $m = [K:F]$ )

Consider a system

$$\begin{array}{l} \sigma_1(w_1)x_1 + \sigma_2(w_1)x_2 + \dots + \sigma_n(w_1)x_n = 0 \\ \vdots \\ \sigma_1(w_m)x_1 + \sigma_2(w_m)x_2 + \dots + \sigma_n(w_m)x_n = 0 \end{array} \quad \left( \text{or } \left( \sigma_j(w_i) \right)_{i,j} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0 \right)$$

By the assumption,  $n > m$ , so the system has a non-trivial solution  $\beta_1, \dots, \beta_n$  in  $K$ .

(By the dimension theorem, nullity  $A = n - \text{rank } A \geq n - m > 0$ .)

Given  $a_1, \dots, a_m \in F$ , these are fixed by  $\sigma_i \in G$ , by the definition of  $F$ .

Multiplying  $a_i$  to the  $i$ th row of the equation, then

$$\sigma_1(a_1 w_1) \beta_1 + \dots + \sigma_n(a_1 w_1) \beta_n = 0$$

$$\sigma_1(a_m w_m) \beta_1 + \dots + \sigma_n(a_m w_m) \beta_n = 0.$$

So, adding all the equations  $i$  for  $\alpha = a_1 w_1 + \dots + a_m w_m$ ,

$$\beta_1 \sigma_1(\alpha) + \dots + \beta_n \sigma_n(\alpha) = 0.$$

Since  $a_1, \dots, a_m$  was arbitrary and  $w_1, \dots, w_m$  forms basis for  $K$  over  $F$ , the  $\alpha$  is arbitrary element of  $K$ . Moreover, since  $\beta_1, \dots, \beta_n$  are not all zero, it contradicts the  $G$  is linearly independent, by the Dedekind Character Theorem. Now, suppose that  $n < [K:F]$ . That is, there are more than  $n$   $F$ -linearly independent elements, say  $d_1, \dots, d_{n+1}$ . Consider a system of  $n+1$  unknowns,

$$\sigma_1(d_1)x_1 + \dots + \sigma_1(d_{n+1})x_{n+1} = 0$$

$$\sigma_n(d_1)x_1 + \dots + \sigma_n(d_{n+1})x_{n+1} = 0. \text{ Then, it has } n \text{ equation and } n+1 \text{ unknowns}$$

So there is non-trivial solution  $\beta_1, \dots, \beta_{n+1}$  in  $K$ .

Moreover, since  $\sigma_1 = \text{id}$  and  $d_1, \dots, d_{n+1}$  are  $F$ -linearly independent, not all  $\beta_i$  lies in  $F$ .

Corollary. Let  $K/F$  be any finite extension. Then,

$$|\text{Aut}(K/F)| \leq [K:F]$$

In Particular, the equality holds if and only if  $F = K^{\text{Aut}(K/F)}$   
if and only if  $K/F$  is Galois extension  
if and only if  $K$  is splitting field of separable polynomial over  $F$ .

Proof.

Corollary.

Let  $G$  be a finite subgroup of  $\text{Aut}(K)$ . Then

$$\text{Aut}(K/K^G) = G.$$

And, if  $G_1 \neq G_2$ , then  $K^{G_1} \neq K^{G_2}$ .

# Fundamental Theorem of Galois Theory.

Let  $K/F$  be a Galois Extension. Denote  $G = \text{Gal}(K/F)$ .

1). A map

$$\mathcal{Q}: \{H \leq \text{Gal}(K/F) \mid H \text{ is subgroup}\} \rightarrow \{F \leq E \leq K \mid E \text{ is intermediate field}\}$$

$$: H \mapsto K^H$$

is bijection, with the inverse  $\mathcal{Q}^{-1}(E) = \text{Gal}(K/E)$ .

2). Given  $H_1, H_2 \leq G$ ,

$$H_1 \leq H_2 \text{ if and only if } K^{H_2} \leq K^{H_1}.$$

3). Given  $H \leq G$ ,

$$[K:K^H] = |H| \quad \text{and} \quad [K^H:F] = |\text{Gal}(K/F):H|$$

$$\begin{array}{c} K \\ / \quad | \quad [H] \\ |G| \quad K^H \\ \backslash \quad | \quad [G:H] \\ F \end{array}$$

4). Given  $H \leq G$ ,  $K/K^H$  is always Galois, with  $\text{Gal}(K/K^H) = H$ .

Moreover,  $K^H/F$  is Galois if and only if  $H \leq G$  is normal.

If  $K^H/F$  is Galois, then  $\text{Gal}(K^H/F) \cong \text{Gal}(K/F)/H$ .

Further, there exists one-to-one correspondence between

$$\{\mathcal{Q}: K^H \rightarrow \overline{K} : \mathcal{Q} \text{ is embedding which fix } F\} \leftrightarrow \{\sigma H \mid \sigma \in G\},$$

5). Given  $H_1, H_2 \leq G$ , then

$$K^{\langle H_1, H_2 \rangle} = K^{H_1} \cap K^{H_2} \quad \text{and} \quad K^{H_1 H_2} = K^{H_1} K^{H_2}$$

